

AADIT AGRAWAL

Codex Consultant · BTech CCE '27 · AI/ML · SWEng · HPC · DevOps
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EDUCATION

Manipal Institute of Technology, MAHE

Jul 2023 - May 2027

BTech - Computer and Communication Engineering

EXPERIENCE

Codex Consultant · OpenAI India

May 2026 - Present

Head Student Researcher · CASIR, MAHE - ecitis.org

May 2025 - Present

- Architected on-prem inference/HPC stack serving 1,500+ daily requests at 99.2% uptime on SGLang, vLLM, and llama.cpp.
- Configured a first-of-its-kind-in-India 4x Mac Studio cluster (512GB unified memory, 16TB storage) with Thunderbolt RDMA / InfiniBand-style fabric for MLX/MPS inference and training; leading USD 300K NVIDIA Blackwell bring-up.
- Benchmarking PyTorch, JAX, vLLM, SGLang, TensorRT-LLM, CUDA, NCCL, and MLX runtimes for heterogeneous on-prem inference and training workloads.
- Delivered production AI systems through CASIR's Schneider Electric collaboration: PLC code translation (70% migration-effort reduction), retrieval over 2,000+ technical documents (94% accuracy, 3x faster), and application-security analysis (3 weeks to 3 days).
- Built real-time computer vision on NVIDIA Jetson + DJI drone at 30fps edge inference; 2 peer-reviewed publications in progress.

Web and Design Engineer · Rossmann New Media 2 Inc.

Jan 2026 - Present

- Freelance/contract role redesigning and redeveloping rossmanngroup.com: WordPress to Next.js rebuild, responsive UI, performance fixes, and production deployment support.
- Technical SEO for data-recovery queries moved primary rankings from 100+ to the top 5; earlier 2022-2024 work included thumbnails, posts, and merchandise for million+ view media.

DevOps/MLOps Engineering Intern · Pixa AI, Bengaluru

Dec 2025

- Provisioned H100/H200/B200 GPU compute and CI/CD pipelines for Phi-4 TTS/STT fine-tuning; supported 2 Model API releases at 99.5% deployment success.
- Debugged distributed training and deployment issues across NCCL, drivers, containers, and runtime configuration.

RESEARCH / WRITING

FL + Digital Twin Route Optimisation · SCE, MAHE

Jun 2024 - Jul 2025

- Built GPS telemetry pipeline over 500K+ datapoints and dashcam CV dataset with OpenCV + YOLO over 12,000+ labelled samples.
- Implemented privacy-preserving federated-learning pipeline for distributed route optimisation; evaluated client drift and non-IID convergence behaviour.

ECITIS Technical Blog · blog.ecitis.org

2025 - Present

- Authored 18 long-form guides on TensorRT/Triton serving, Apple MLX/Metal, ROCm, RLHF/DPO, MoE, edge AI, VLMs, coding agents, and HPC cluster tuning.

PROJECTS

Homelab Infrastructure · Self-hosted cloud

2022 - Present

- Operate 70+ services on Proxmox/Docker with 98% uptime, 8TB MergerFS, Grafana/Prometheus/Loki monitoring, 3-2-1 backups, self-hosted git/blog/design services, and GPU media/AI workloads.
- Fine-tuned/deployed object-detection models on security-camera feeds at 30fps and offloaded 95% of 4K media/AI compute from CPU to GPU.
- Mitigated live DDoS traffic at 50K+ req/min using fail2ban, geo-blocking, segmentation, mTLS, and auth proxies; reduced attack surface by 95%.

The TIK Menu · Campus food platform - fc2.coolstuff.work

2025 - Present

- Built zero-cost campus menu platform with 5,000+ DAU, 150K+ monthly requests, dynamic XLSX parsing, Cloudflare R2 storage, REST API, PWA/offline support, and campus-wide adoption in under 7 days.

Coolstuff.work + OSS Tooling · git.aadit.cc/aadit

2023 - Present

- Built utility suite reaching 10,000+ aggregate DAU: OR-Free model/pricing tracker, MIT Courses Search (500+ graph nodes), Campus Directory, Contact Cards, Mass Contact Maker, and transcript-cli.
- Maintained mostly zero-cost deployments across Cloudflare, self-hosted Git, and static/PWA surfaces with fast iteration loops for campus-scale tools.

Agent + Developer Tooling · OSS / self-hosted

2024 - Present

- Published agentskills, a t3code fork, and ophoner mobile agent loop; built tmux + Android shell workflows for parallelised coding-agent experiments.
- Built Typsmthng, a Typst editor with WASM preview and vim-mode experiments; run self-hosted CI/sandboxing and CodeRabbit-backed review flows for public repos.

LEADERSHIP

Founding President · PRISM

Jan 2026 - Present

- Founded 150+ member interdisciplinary student research club; serve on MAHE Academic Advisory Board for AI Curriculum and Adoption.
- Coordinate research groups across AI, systems, product, and design; maintain PRISM's web presence, GitHub organisation, and student-facing technical material.

Student Society Leadership · MIST, Hult Prize, Astronomy Club, TechTatva

2023 - 2025

- Led student security, design, and event work: CTF infra for 300+ users at 100% uptime, 12+ workshops for 150+ participants, Hult Prize graphics, Astronomy Club outreach, and TechTatva CAP organizing.
- Built club web properties, security-awareness material, and event communications; competed in national CTFs with top-10% finishes.

SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, Go, Rust, Zig, Bash, SQL, Java

AI/ML & HPC: PyTorch, JAX, Transformers, SGLang, vLLM, llama.cpp, TensorRT-LLM, MLX, CUDA, NCCL, OpenCV, YOLO

Infrastructure: Linux, Docker, Proxmox, GitHub Actions, Terraform, Ansible, Nginx, Caddy, Tailscale, Grafana, Prometheus, Cloudflare

Web & Data: Next.js, React, Astro, Svelte, Tailwind, Node.js, Bun, FastAPI, tRPC, PostgreSQL, Redis, pgvector

Agent Tooling: Codex, Claude Code, Pi, Droid, Cursor, opencode, MCP, CodeRabbit